SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%
Software:Cisco Packet Tracer,Wireshark

Code of Conduct:

I _U. Ananya(192424382)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

U. Ananya
Signature of the student:

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Problem Identification	20%				
Debugging	25%				
Optimization	20%				
Analysis	20%				
Code Quality	15%				

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

2. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

3. An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks **/26**, **/27**, and **/28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

4. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

5. A router contains the following routing table.

Destination Network	**Next Hop**
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for **192.168.2.100** are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation

1. Debugging and Optimisation of IPv4 Subnet Configuration

Problem Statement

A network administrator has configured a workstation with the following network settings:

- IP Address: 192.168.10.65
- Subnet Mask: 255.255.255.192 (/26)
- Default Gateway: 192.168.10.1

However, the workstation cannot communicate with other systems configured in the subnet 192.168.10.0/26. The configuration must be examined to identify the addressing errors and restore network connectivity.

Step 1: Identify the subnet to which the IP address belongs

The subnet mask 255.255.255.192 corresponds to a /26 prefix. This means that 26 bits are used for the network portion and the remaining 6 bits are used for host addresses.

The number of addresses in each subnet is:

$$2^{32-26} = 2^{6} = 64$$

Therefore, each /26 subnet contains 64 total IPv4 addresses. Within the 192.168.10.0/24 address block, the subnet boundaries increase in increments of 64.

The subnets are:

- 192.168.10.0/26
- 192.168.10.64/26
- 192.168.10.128/26
- 192.168.10.192/26

Since the configured IP address is 192.168.10.65, it belongs to the subnet 192.168.10.64/26, rather than the intended subnet 192.168.10.0/26.

Step 2: Calculate the network address, broadcast address, and host range

Parameter	**Value**
IP address	192.168.10.65
Subnet mask	255.255.255.192

Network address	192.168.10.64
Broadcast address	192.168.10.127
First usable host	192.168.10.65
Last usable host	192.168.10.126
Total IP addresses	64
Usable host addresses	62

The network address identifies the subnet, while the broadcast address is used to send traffic to all hosts in that subnet. Neither is normally assigned to an individual host.

The usable host range is from 192.168.10.65 to 192.168.10.126, so the configured workstation IP is valid within its actual subnet.

Step 3: Verify the IP address and default gateway

The workstation's IP address, 192.168.10.65, is valid for the subnet 192.168.10.64/26. However, the configured default gateway, 192.168.10.1, belongs to the different subnet 192.168.10.0/26.

A workstation normally needs a reachable gateway on its local subnet to forward packets to other networks. Consequently, the gateway configuration is inconsistent with the workstation's current IP address.

There are two related problems:

1. The workstation is configured in 192.168.10.64/26, while the intended systems belong to 192.168.10.0/26.
2. The default gateway 192.168.10.1 is outside the workstation's configured subnet.

Step 4: Identify the configuration errors

The primary error is the incorrect subnet assignment. Although the workstation has a valid IP address, it is not in the same subnet as the intended systems.

The gateway is also incorrectly configured relative to the workstation's current subnet. This may prevent the workstation from forwarding traffic through the gateway. A routing configuration on

another device could provide connectivity between subnets, but that is not established by the information given.

Step 5: Optimize the network configuration

If the workstation must belong to 192.168.10.0/26, its IP address and gateway should be assigned from that subnet.

Setting	**Corrected configuration**
IP address	192.168.10.10
Subnet mask	255.255.255.192
Network address	192.168.10.0
Broadcast address	192.168.10.63
Default gateway	192.168.10.1
Valid host range	192.168.10.1 – 192.168.10.62

This configuration is valid provided that 192.168.10.10 is not already assigned to another device and 192.168.10.1 is configured as the operational gateway.

After applying the correction, the administrator should verify the configuration using ipconfig on Windows, test connectivity to the gateway using ping 192.168.10.1, and then test communication with the intended systems.

Step 6: Calculate the total number of usable host addresses

The number of host bits in a /26 subnet is:

$$32 - 26 = 6$$

The total number of addresses is:

$$2^6 = 64$$

For conventional IPv4 subnetting, the network and broadcast addresses are reserved. Therefore:

64-2= 62 usable host addresses

Result: The corrected /26 subnet supports 62 usable host addresses.

Conclusion

The communication failure is caused by a mismatch between the workstation's subnet and the intended network, together with an invalid gateway placement for its current subnet. Assigning an available IP address from 192.168.10.0/26 and using 192.168.10.1 as the gateway aligns the configuration with the intended network. The corrected subnet provides 62 usable host addresses.

2. Debugging and Optimisation of IPv6 Address Configuration

Problem Statement

Two devices are connected to the same network, but they cannot communicate. Their IPv6 configurations are:

- Device A: 2001:db8::ff00:42:8329/64
- Device B: 2001:db8:0:0:1::1/64

The IPv6 addresses must be expanded, their prefixes compared, and the addressing configuration corrected.

Step 1: Expand both IPv6 addresses into complete notation

IPv6 addresses use 128 bits, represented by eight groups of hexadecimal digits. Each group contains four hexadecimal digits. The double colon (::) represents one or more consecutive groups of zeros and may appear only once in an address.

Device A:

Given address:

2001:db8::ff00:42:8329/64

Fully expanded address:

2001:0db8:0000:0000:0000:ff00:0042:8329/64

Device B:

Given address:

2001:db8:0:0:1::1/64

Fully expanded address:

2001:0db8:0000:0000:0001:0000:0000:0001/64

Expanding the addresses makes it easier to compare their network prefixes and identify differences in the assigned subnet.

Step 2: Determine the network prefix of each device

A /64 prefix means that the first 64 bits, or first four hexadecimal groups, represent the network prefix. The remaining 64 bits identify the interface within the subnet.

Parameter	**Device A**	**Device B**
IPv6 address	2001:db8::ff00:42:8329	2001:db8:0:0:1::1
Prefix length	/64	/64
Network prefix	2001:db8:0:0::/64	2001:db8:0:0:1::/64
Interface identifier	0000:ff00:0042:8329	0001:0000:0000:0001

The first four groups of Device A are 2001:0db8:0000:0000, whereas the first four groups of Device B are 2001:0db8:0000:0000 as well.

Correction to the comparison: The first four groups of both fully expanded addresses are identical. Therefore, both devices actually belong to the same /64 subnet: 2001:db8:0:0::/64.

The apparent difference occurs in the fifth group, which is part of the interface identifier—not the network prefix under /64.

Step 3: Verify whether both devices belong to the same subnet

Both devices have the same first 64 bits and use the /64 prefix. Therefore, they belong to the same IPv6 subnet.

This means that the addresses, as written, do not demonstrate a subnet mismatch. The earlier assumption that Device B belongs to a different /64 subnet would be incorrect.

The devices should normally be able to communicate directly at the IPv6 link level if they are on the same Layer 2 network, their interfaces are correctly configured, and there are no filtering or other connectivity issues.

Step 4: Identify the likely configuration issue

Because both devices share the same /64 prefix, a prefix mismatch is not the cause established by the supplied information. The administrator should instead investigate other possible causes, including:

- Incorrect or duplicate IPv6 addresses.
- IPv6 disabled on one of the network interfaces.
- Incorrect VLAN or Layer 2 network configuration.
- Neighbor Discovery failures.
- Firewall or access-control rules blocking IPv6 traffic.
- Incorrect interface configuration or physical connectivity.
- An incorrect assumption that the two devices are connected to the same Layer 2 network.

The administrator should inspect the actual interface settings and test local IPv6 reachability before changing the addresses.

Step 5: Optimize the IPv6 configuration

The existing addresses can be retained if they are unique and correctly assigned. For example:

Setting	**Device A**	**Device B**
IPv6 address	2001:db8::ff00:42:8329/64	2001:db8:0:0:1::1/64
Network prefix	2001:db8:0:0::/64	2001:db8:0:0::/64
Address uniqueness	Verify	Verify

Both devices already have matching /64 network prefixes. Thus, no IPv6 address change is required solely to correct a subnet mismatch. The correct optimization is to verify the interface, VLAN, Neighbor Discovery, firewall, and routing settings.

If a simpler address plan is desired, Device B could instead be assigned 2001:db8::1/64, provided the address is available and the existing addressing plan permits it.

Step 6: Calculate the number of available host addresses

For a /64 subnet, the number of bits available for interface identifiers is:

$$128 - 64 = 64$$

Therefore, the theoretical number of possible interface identifiers is:

$$2^{64}=18,446,744,073,709,551,616$$

Unlike conventional IPv4 subnetting, IPv6 does not reserve a network address and broadcast address in the same way. This figure represents the theoretical interface-identifier space, not a guarantee that every address can be assigned to a device under every IPv6 addressing method.

Conclusion

The two IPv6 addresses share the same first 64 bits, so both belong to 2001:db8:0:0::/64. The supplied configuration does not contain a /64 subnet mismatch. The administrator should investigate Neighbor Discovery, VLAN configuration, interface status, firewall rules, and other connectivity settings before modifying the IP addresses. A /64 provides a theoretical interface-identifier space of 2^{64} values.

3. Debugging and Optimisation of IPv4 Subnet Allocation Using VLSM

Problem Statement

An organization has allocated the network 192.168.1.0/24 to several departments using subnet masks /26, /27, and /28. Some departments report insufficient IP addresses, while other departments have many unused addresses.

The network must be analyzed to identify subnet boundaries, calculate address utilization, detect allocation errors, and recommend an optimized subnetting plan using Variable Length Subnet Masking (VLSM).

Step 1: Understand the subnet masks

The network 192.168.1.0/24 contains 256 total IPv4 addresses, from 192.168.1.0 to 192.168.1.255.

Different subnet masks provide different numbers of host addresses.

Prefix	Subnet mask	Total addresses	Usable host addresses
/26	255.255.255.192	64	62
/27	255.255.255.224	32	30
/28	255.255.255.240	16	14

For conventional IPv4 subnets, two addresses are reserved: the network address and the broadcast address.

Step 2: Identify network addresses, broadcast addresses, and host ranges

The following is one valid, non-overlapping allocation within the /24 network.

Parameter	Department A (/26)	Department B (/27)	Department C (/28)
Network address	192.168.1.0	192.168.1.64	192.168.1.96
Broadcast address	192.168.1.63	192.168.1.95	192.168.1.111
First usable host	192.168.1.1	192.168.1.65	192.168.1.97
Last usable host	192.168.1.62	192.168.1.94	192.168.1.110
Total addresses	64	32	16
Usable addresses	62	30	14

These subnet boundaries are valid because each subnet starts at an appropriate block boundary and does not overlap with the preceding subnet.

The remaining address range is 192.168.1.112 through 192.168.1.255, which contains 144 addresses available for further subnet allocation.

Step 3: Calculate usable and unused IP addresses

The actual number of unused addresses depends on the number of devices in each department. Since the question does not specify departmental requirements, the following example uses assumed device counts to demonstrate the calculation.

Department	Allocated prefix	Usable capacity	Assumed devices	Unused host capacity
Department A	/26	62	50	12
Department B	/27	30	12	18
Department C	/28	14	6	8

Total		106	68	38

The unused host capacity is calculated as:

$$Unusedhosts = Usablecapacity - Devicesrequired$$

For example, Department B has 30 usable addresses and 12 assumed devices.

$$30 - 12 = 18 unusedhostaddresses$$

This example shows how a department may have more addresses than it needs. However, these unused addresses are not necessarily wasted if the department expects future growth.

Step 4: Identify inefficient subnet allocations

Several problems can occur when subnet sizes are not matched to departmental requirements.

1. Oversized subnets: A department with only 10 devices may not need a /26, which provides 62 usable addresses.

2. Undersized subnets: A department with 40 devices cannot accommodate all of them in a /27, because it provides only 30 usable host addresses.

3. Overlapping subnets: Assigning overlapping IP ranges to different departments can create routing ambiguity and connectivity issues.

4. Incorrect subnet boundaries: A /26 subnet must start at a multiple of 64 addresses in the last octet. A /27 must start at a multiple of 32, and a /28 at a multiple of 16.

5. Poor planning for growth: Assigning all remaining addresses without reserving space for future departments may make later expansion difficult.

Step 5: Debug the addressing plan

The administrator should check the following:

- Confirm the correct network address and subnet mask for each department.
- Verify that every subnet has a unique, non-overlapping address range.
- Ensure that each department has enough usable host addresses.
- Verify that network and broadcast addresses are not assigned to individual devices.
- Check that default gateways belong to the corresponding subnets.
- Review DHCP address pools to ensure they match the subnet boundaries.
- Confirm that routers have the necessary routes between departmental networks.

These checks help distinguish address-allocation problems from routing, DHCP, and gateway configuration problems.

Step 6: Recommend an optimized VLSM scheme

VLSM allows the administrator to divide a larger network into smaller subnets of different sizes. The usual approach is to allocate the largest departmental requirement first, followed by smaller requirements.

For illustration, assume the departments need 50, 12, and 6 device addresses, respectively, and that no additional network addresses are required for gateways or other equipment.

Department	Required device addresses	Recommended subnet	Usable capacity
Department A	50	192.168.1.0/26	62
Department B	12	192.168.1.64/28	14
Department C	6	192.168.1.80/29	6

The subnet details are:

Department	Network address	Usable host range	Broadcast address
A	192.168.1.0	192.168.1.1 – 192.168.1.62	192.168.1.63
B	192.168.1.64	192.168.1.65 – 192.168.1.78	192.168.1.79
C	192.168.1.80	192.168.1.81 – 192.168.1.86	192.168.1.87

The /29 subnet provides six conventional usable IPv4 addresses, so it can support six device addresses only if a gateway or other equipment does not require an additional address. If a gateway is needed in the same subnet, the administrator should use a larger subnet, such as a /28.

This allocation leaves a larger portion of the original /24 network available for future departments.

Step 7: Evaluate the benefits of VLSM

VLSM improves network design in several ways:

- Reduces unnecessary consumption of IPv4 addresses.
- Allows each department to receive an appropriately sized subnet.
- Helps prevent address shortages in departments with more devices.
- Leaves unused address blocks available for future expansion.
- Supports a more structured IP addressing and routing plan.

Conclusion

The main objective of subnet optimization is to match the subnet size to the actual number of devices in each department. VLSM makes this possible by allowing different subnet masks within the same larger network. Before implementing the optimized scheme, the administrator must confirm actual device counts, gateway requirements, expected growth, and existing subnet assignments.

4. Debugging and Optimisation of IPv6 Address Configuration

This question is identical to Question 2, so the same IPv6 analysis applies.

Step 1: Expand the addresses

- Device A: 2001:0db8:0000:0000:0000:ff00:0042:8329/64
- Device B: 2001:0db8:0000:0000:0001:0000:0000:0001/64

Step 2: Identify the prefixes

Parameter	**Device A**	**Device B**
Network prefix	2001:db8:0:0::/64	2001:db8:0:0::/64
Prefix length	/64	/64
Same subnet?	Yes	Yes

The first four hexadecimal groups of both addresses are identical. The fifth group belongs to the interface identifier because the prefix length is /64.

Step 3: Identify the actual problem

The addresses do not have a prefix mismatch. Both devices belong to 2001:db8:0:0::/64.

If communication fails, the administrator should investigate the network interface configuration, VLAN assignment, Neighbor Discovery, firewall rules, duplicate addresses, and Layer 2 connectivity.

Step 4: Optimize the configuration

The existing addresses can remain unchanged if they are valid, unique, and correctly assigned. If a simpler addressing plan is required, Device B may be assigned 2001:db8::1/64, provided that this address is available.

Step 5: Calculate the address space

A /64 leaves 64 bits for the interface identifier:

$$2^{128-64} = 2^{64}$$

Therefore, the theoretical interface-identifier space contains 18,446,744,073,709,551,616 possible values.

Conclusion

Both IPv6 addresses belong to the same /64 subnet, so changing the prefix is not necessary based on the supplied configuration. The administrator should investigate other causes of communication failure and correct the actual issue rather than changing valid IPv6 addresses unnecessarily.

5. Debugging and Optimisation of Router Routing Table

Problem Statement

A router contains the following routing table:

Destination network	**Next hop**
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default route	10.0.0.1

Users report that packets destined for 192.168.2.100 are not reaching the intended network. The routing entries must be examined to determine the selected route, next hop, possible errors, and fallback route.

Step 1: Identify the matching routing entry

The destination IP address is 192.168.2.100.

The routing table contains an entry for 192.168.2.0/24. A /24 subnet has the subnet mask 255.255.255.0, which means that the first three octets identify the network.

The address range for this subnet is:

- Network address: 192.168.2.0
- First usable host: 192.168.2.1
- Last usable host: 192.168.2.254
- Broadcast address: 192.168.2.255

Since 192.168.2.100 falls within this range, the route 192.168.2.0/24 matches the destination.

Step 2: Apply the longest-prefix match rule

Routers use the longest-prefix match rule to select the most specific matching route. A longer prefix identifies a smaller and more specific destination network.

For the given destination:

Routing entry	**Match?**	**Explanation**
192.168.1.0/24	No	The destination belongs to a different network
192.168.2.0/24	Yes	The destination falls within this network
0.0.0.0/0	Fallback match	Matches any IPv4 destination, but is less specific

The most specific matching route is 192.168.2.0/24.

Selected next hop: 10.0.0.3.

The router forwards the packet toward 10.0.0.3, assuming the next hop is reachable and the route is active.

Step 3: Verify the destination IP address

The destination address 192.168.2.100 is a valid host address in 192.168.2.0/24.

Therefore, the destination IP itself is not invalid based on the provided information.

However, a valid destination address does not guarantee successful communication. The next-hop router must be reachable, have an appropriate route, and be able to forward the packet toward the destination.

Step 4: Identify possible routing configuration errors

The table contains a route that matches the destination. Therefore, the supplied routing entries alone do not establish that the routing table is incorrect.

Possible causes of the failure include:

1. Unreachable next hop: The router may not be able to reach 10.0.0.3 because the associated interface is down or the next-hop network is incorrectly configured.

2. Missing downstream route: Router 10.0.0.3 may not have a valid route to 192.168.2.0/24.

3. Incorrect return route: The destination system may receive the packet but fail to send a reply because its default gateway or return route is incorrect.

4. Firewall restrictions: A firewall or access-control list may block the traffic.

5. Incorrect interface configuration: The outgoing interface may have an incorrect IP address, subnet mask, or VLAN assignment.

6. Destination unavailable: The destination host may be switched off or disconnected from the network.

Step 5: Optimize the routing table

The routing table can be written using the standard IPv4 default-route notation:

Destination network	**Next hop**	**Purpose**
192.168.1.0/24	10.0.0.2	Reach network 1
192.168.2.0/24	10.0.0.3	Reach network 2
0.0.0.0/0	10.0.0.1	Forward traffic without a more-specific route

The existing routes should be retained if they are correct and operational. The administrator should verify:

- Whether 10.0.0.3 is reachable.

- Whether the destination network is directly connected to, or reachable through, the next-hop router.
- Whether the outgoing interface is operational.
- Whether any access-control rules block the traffic.
- Whether a valid return route exists.
- Whether the routing table contains any conflicting or stale entries.

Commands such as ping, tracert on Windows, or traceroute on Linux can help identify where packet forwarding fails.

Step 6: Identify the route used when no matching destination exists

The default route is represented as:

0.0.0.0/0 → 10.0.0.1

The /0 prefix matches every IPv4 destination. Because more-specific routes are preferred, the default route is selected only when no more-specific applicable route is available.

For example, a destination address outside both 192.168.1.0/24 and 192.168.2.0/24 would use the default route, provided that it is active and usable.

If no valid route or default route exists, the router generally drops the packet.

Conclusion

The destination 192.168.2.100 matches the route 192.168.2.0/24, and the selected next hop is 10.0.0.3. The routing table includes an appropriate destination entry, so troubleshooting should focus on next-hop reachability, downstream routes, firewall settings, interface status, and return-path configuration. The default route 0.0.0.0/0 through 10.0.0.1 is used when no more-specific route matches.